

Quantum Mechanics III

Homework 6

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Topic

Identical Particles and Count Correlations

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Identical Particles and Count Correlations. State all assumptions and conventions.

$$[a_i, a_j^\dagger]_{\mp} = \delta_{ij}$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$g^{(2)}(\tau) = \langle : n(t)n(t+\tau) : \rangle / \langle n \rangle^2$$

3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.

$$P_{\text{coin}} = \frac{1}{2}(1 - |\langle \psi_1 | \psi_2 \rangle|^2)$$

Show enough intermediate work that another student can reproduce the calculation.